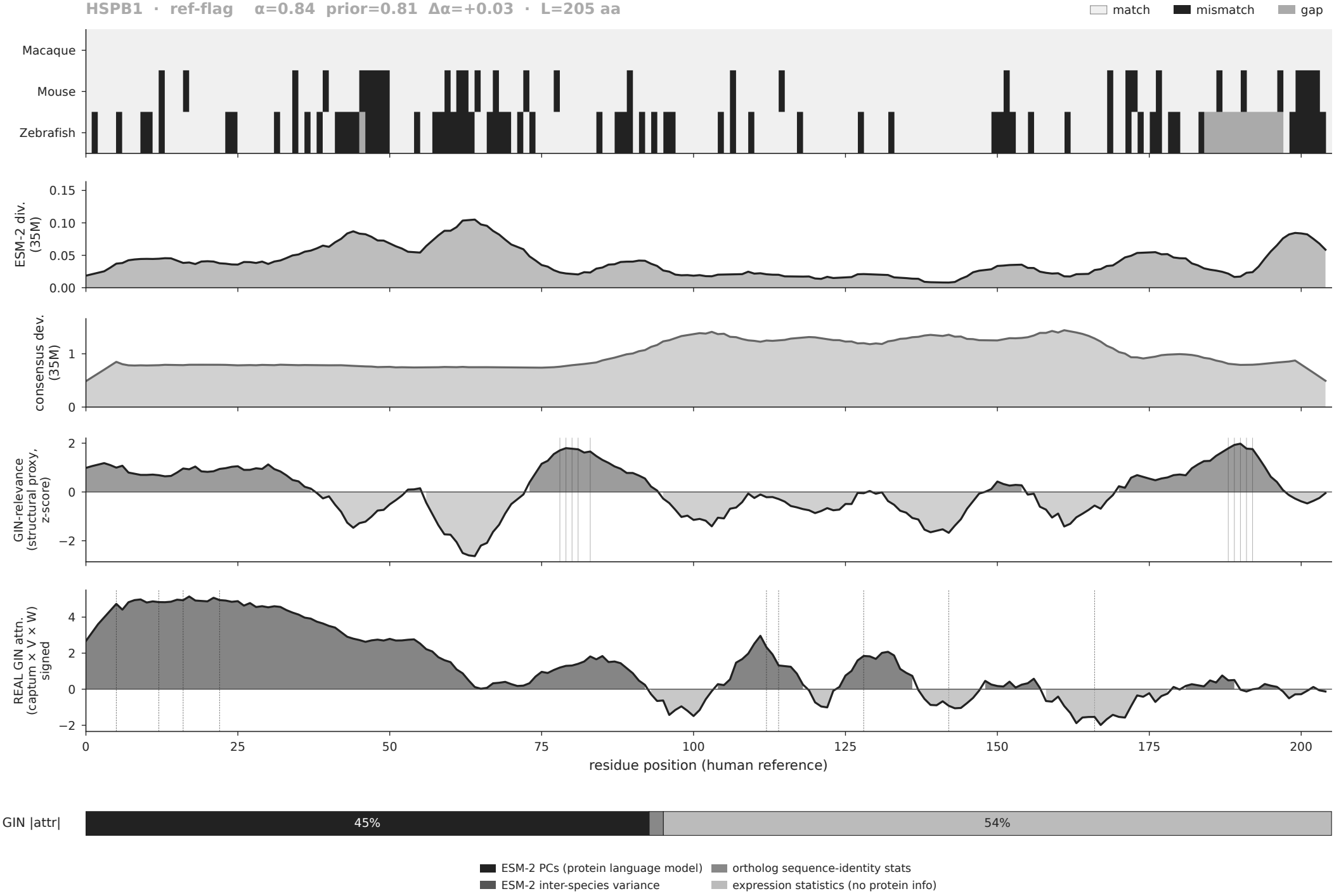
HSPB1 · ref-flag α=0.84 prior=0.81 Δα=+0.03 · L=205 aa
match
mismatch
gap
Macaque
Mouse
Zebrafish
ESM-2 div. (35M)
0.15
0.10
0.05
0.00
consensus dev. (35M)
1
0
GIN-relevance (structural proxy, z-score)
2
0
−2
REAL GIN attn. (captum × V × W) signed
4
2
0
−2
0
25
50
75
100
125
150
175
200
residue position (human reference)
GIN |attr|
45%
54%
ESM-2 PCs (protein language model)
ESM-2 inter-species variance
ortholog sequence-identity stats
expression statistics (no protein info)